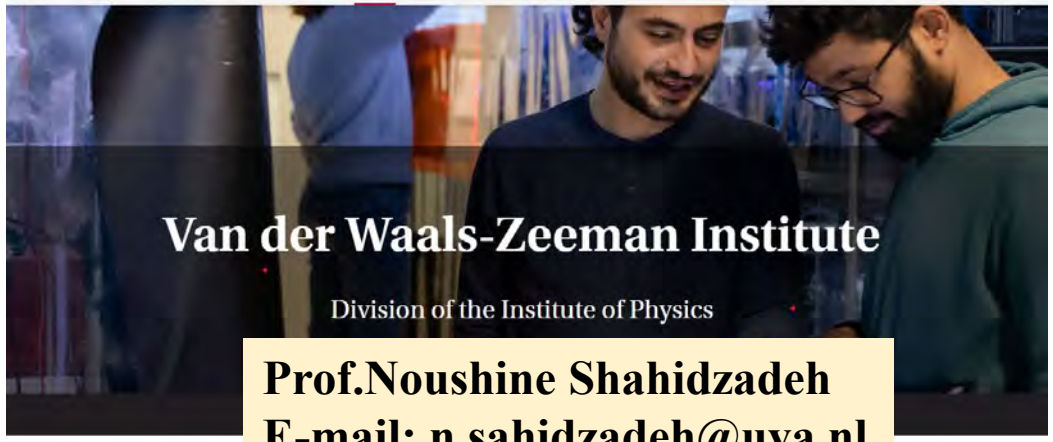
UNIVERSITY OF AMSTERDAM
INSTITUTE OF PHYSICS

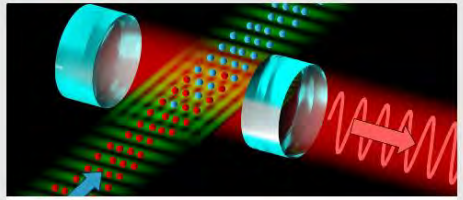
Bachelor Projects 2026



Prof.Noushine Shahidzadeh
E-mail: n.sahidzadeh@uva.nl

At the Van der Waals-Zeeman Institute (WZI) for experimental physics, scientists perform research in the areas of quantum gases & quantum information, quantum materials, and soft matter. The WZI is a division of the Institute of Physics. IoP's contributions to the Advanced Research Center for Nanolithography (ARCNL) are coordinated through WZI.

Research groups at WZI



Quantum Gases & Quantum Information

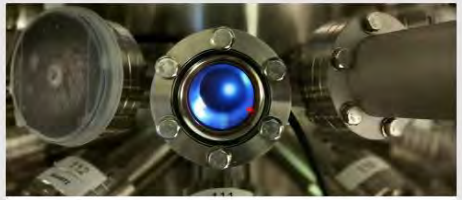
Ultracold atoms and trapped ions enable investigating and utilizing fascinating phenomena in quantum physics.

Quantum Materials

This research group focusses on understanding and utilizing quantum effects in novel materials ranging from nanostructured semiconductors to high temperature superconductors.

Soft Matter

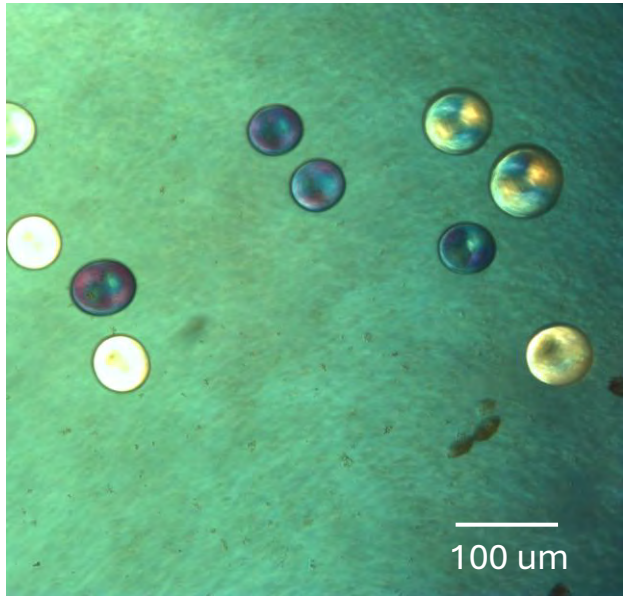
Soft materials have exceptional mechanical, optical or functional properties and find applications in industrial sectors and society.

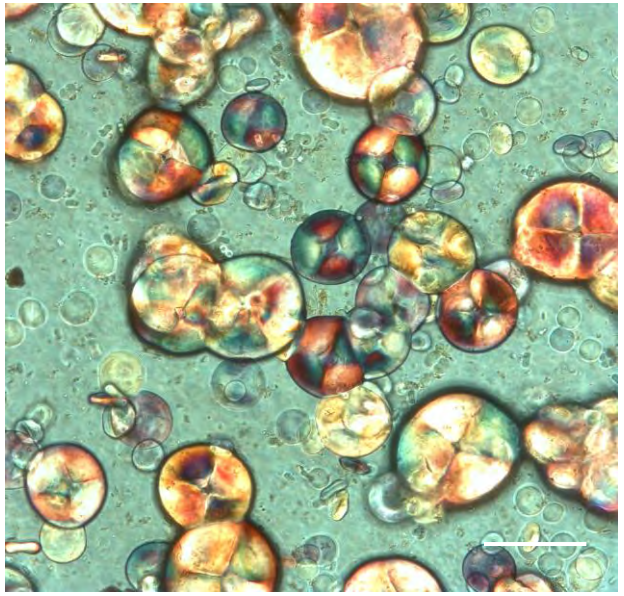


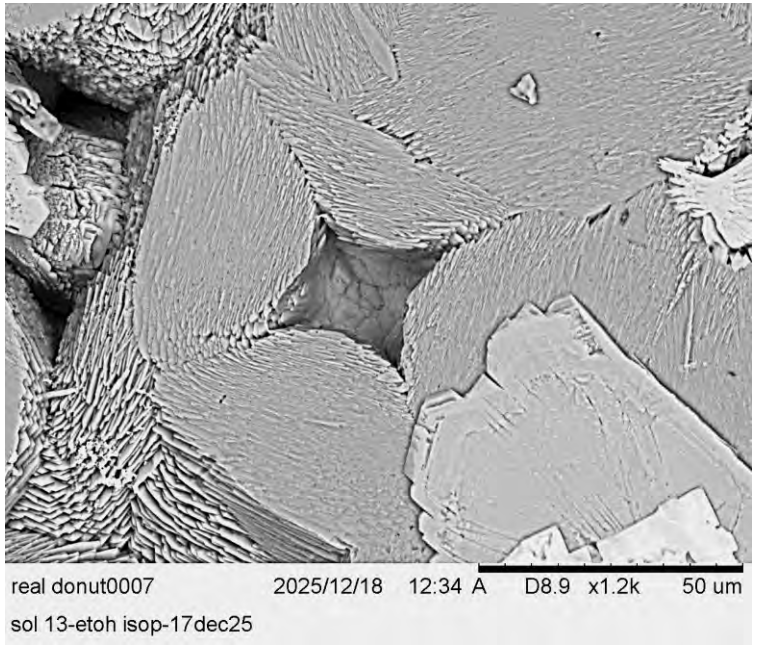
ARCNL

The Advanced Research Center for Nanolithography (ARCNL) is a collaboration between UvA, VU, NWO and ASML. IoP's contributions to ARCNL are coordinated through WZI.

Crystals like donuts!







Context: We found a reproducible method to form Na_2SO_4 crystals in a **very unusual shape** by mixing salts in water (NaCl, $FeSO_4.7H_2O$, Na_2SO_4) and adding a mixture of alcohol. These new donut-shaped crystals sparked our interest since they deviate from the classical Wulff theory that predicts crystals shape. In this project, you will investigate the growth dynamics of this peculiar soft crystals and the role of each component to unveil the mystery behind those rare shapes.

Contact:
Anne Claude: a.claude@uva.nl
Prof. Dr. Noushine Shahidzadeh: N.Shahidzadeh@uva.nl

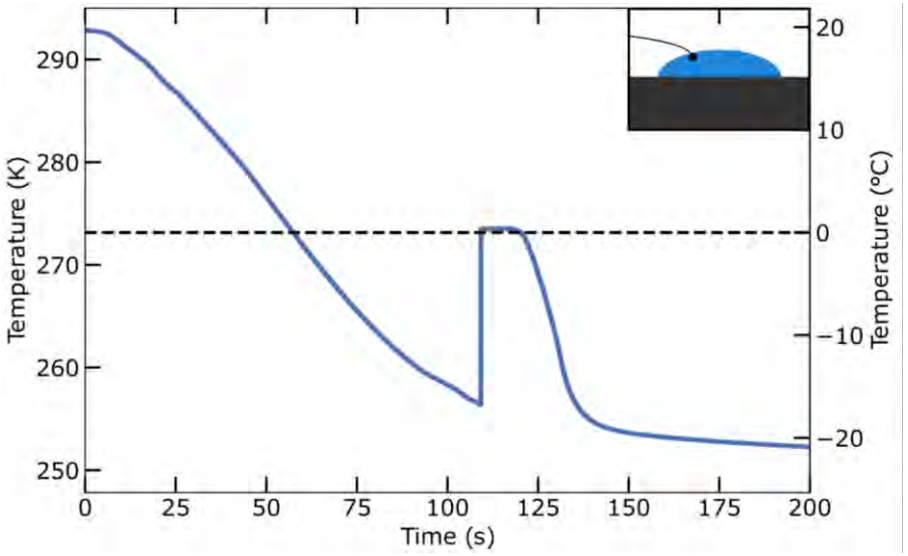
Soft Matter Group, Institute of Physics, UvA

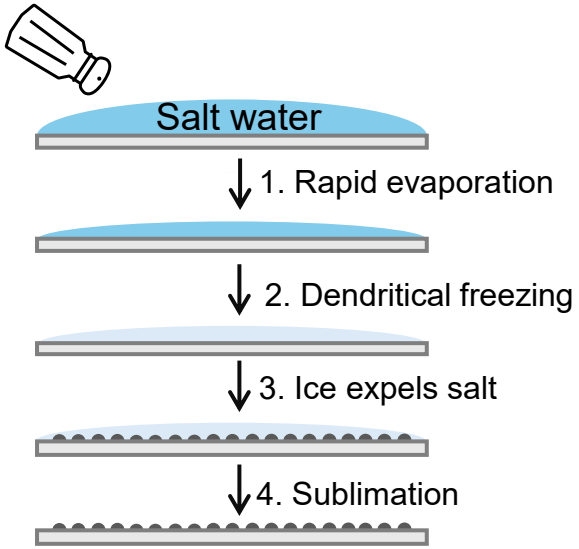
Soft Matter

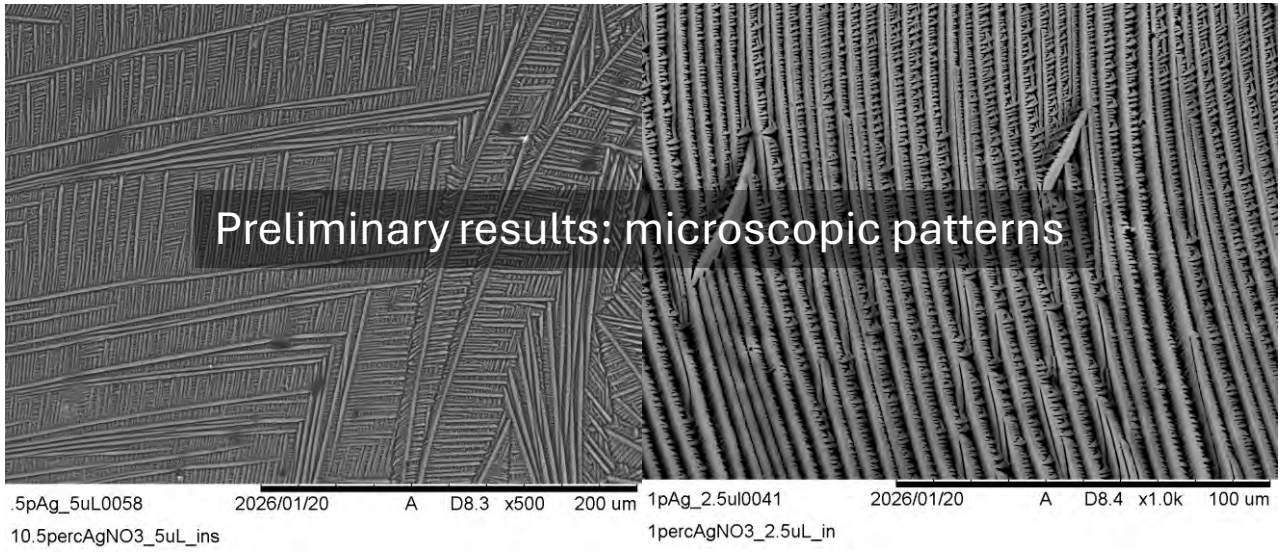
Vacuum Freezing: Ice Templating for Structural Color

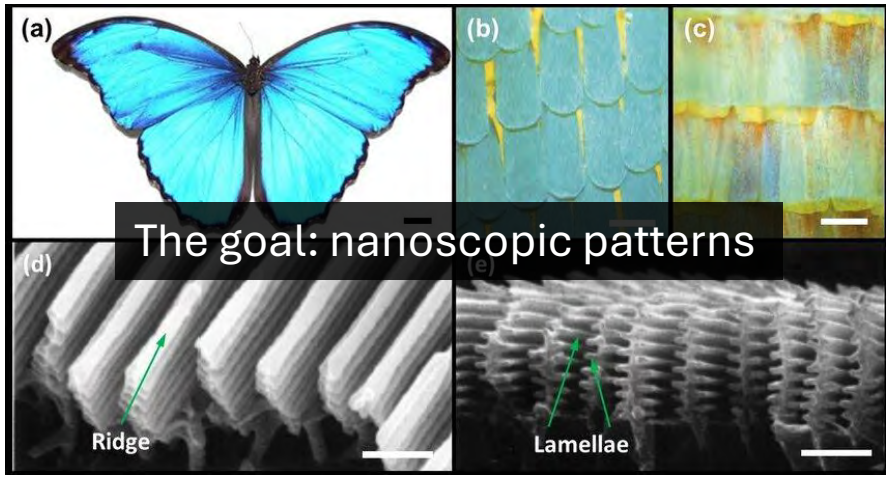
m.demmenie@uva.nl
c.t.vancampenhout@uva.nl

Evaporative cooling in vacuum







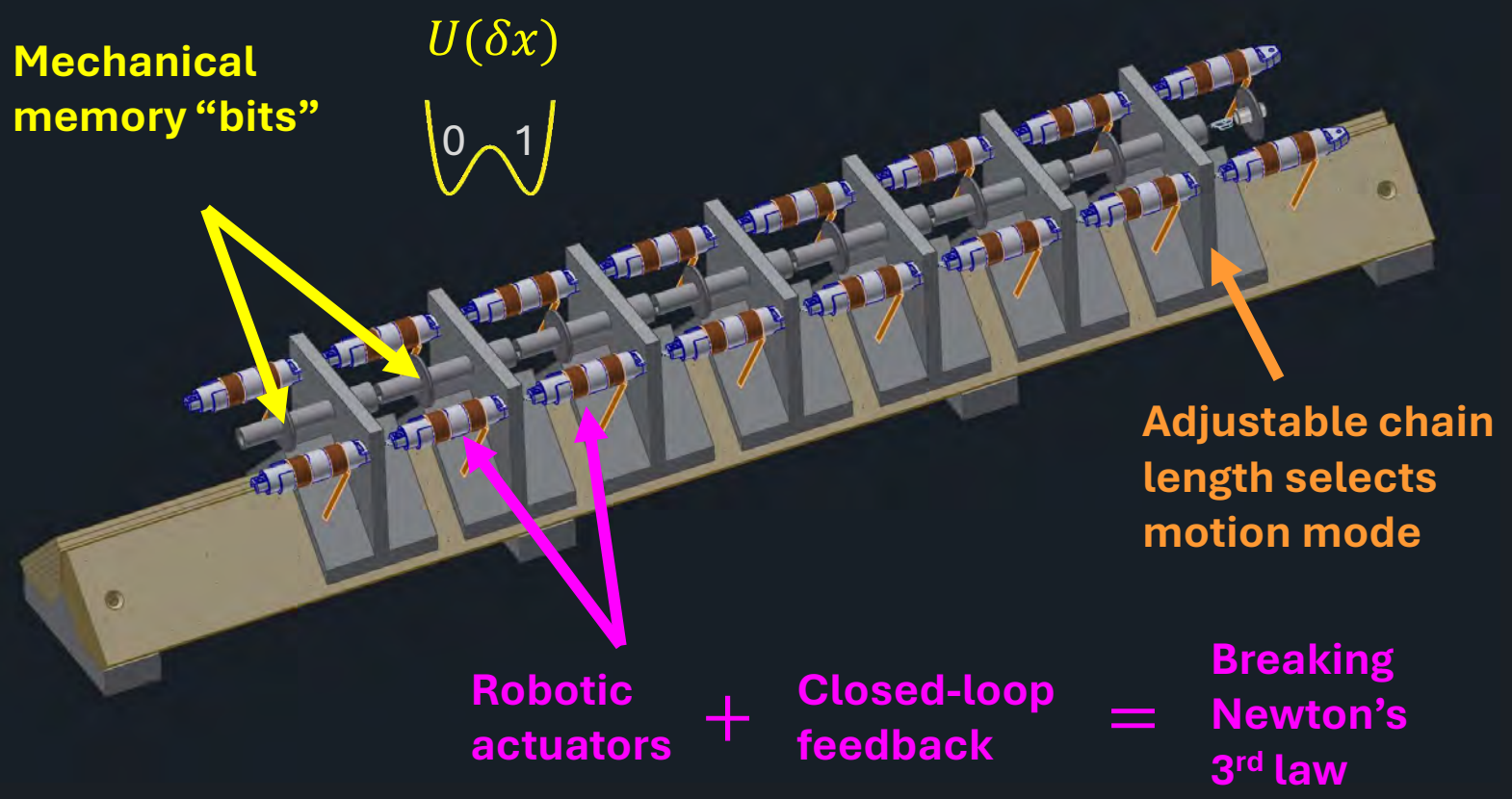


Memory and Dynamic Cycles in Active Multistable Metamaterials

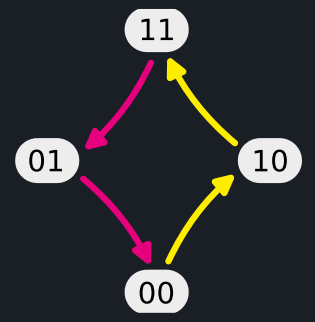
Dor Shohat
Martin van Hecke
Corentin Coulais

Contact: d.shohat@uva.nl

Living organisms can exhibit a range of **coordinated motion patterns** that adapt to a changing environment, yet without central control. How can we build an **autonomous robotic system** that mimics multifunctional motion?



Active cycles emerge from sequences of bit flips!



What you will learn:

- Experimental work with mechanics and electronics
- Tracking and analyzing cycles with image processing
- Theoretical tools to study *dynamical systems* and *memory*

The Physics of Stock Cubes

Contact: Katerina Gkougkousi, PhD candidate
e.a.gkougkousi@uva.nl

How can we control the fracture mechanics of a bouillon (stock) cube?

A bouillon cube usually forms a big blob in our pan, but if we shape it with a geometrical pattern, we can control how that will break.

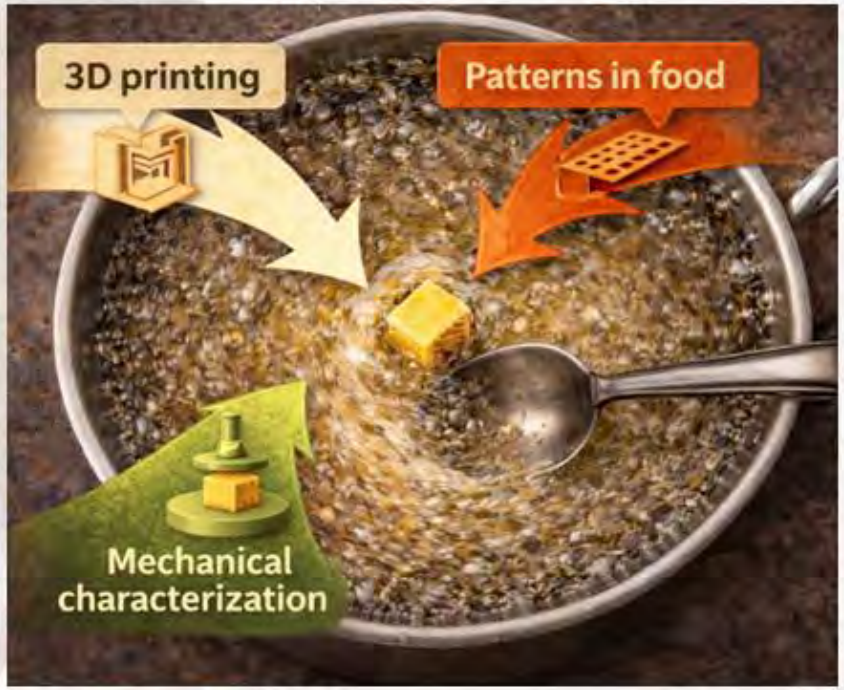
How can we give it such a pattern and what would that mean for its behavior?

Mechanical Testing
+
3D printing

Map for patterned edible materials design, with controlled fracture mechanics

Fat-salt blends with different salt grain size & ratio, just like a bouillon cube !

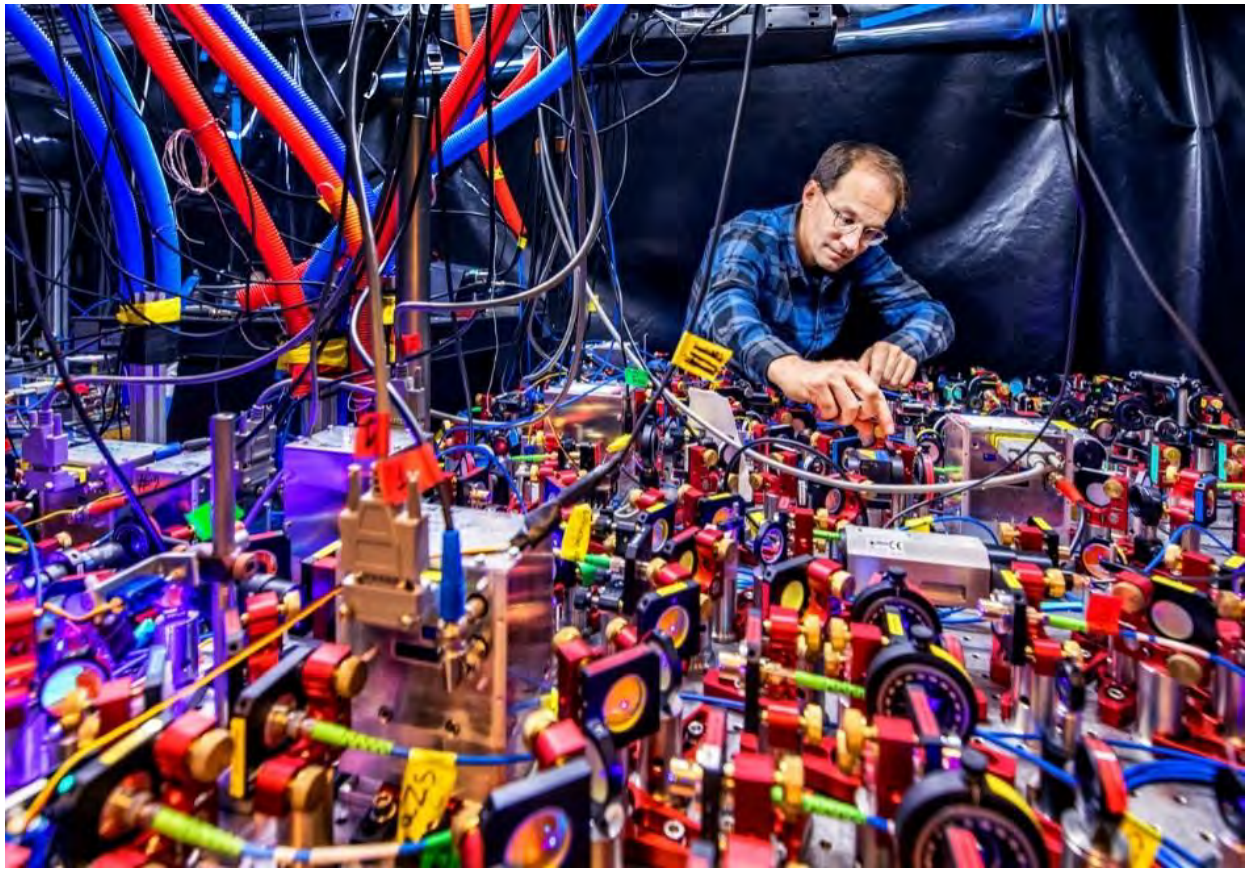
What you'll learn:



Machine Materials Lab
Katerina Gkougkousi
Corentin Coulais

Optical Setup for Ultracold Atom Experiments with Strontium

In ultracold physics we use light to interact with matter at the atomic scale and perform sophisticated experiments. In the SrCAL lab we work with the element strontium, trying to achieve the first continuous atom laser.
In this project we offer hands-on opportunity to dive into this world. You will build an optical setup that plays a crucial role in cooling and trapping strontium atoms. You will work with lasers, atoms, optical fibers and more! Together with the team we will upgrade our already existing setup.



Skills at play: laser physics, atomic physics, laser cooling.

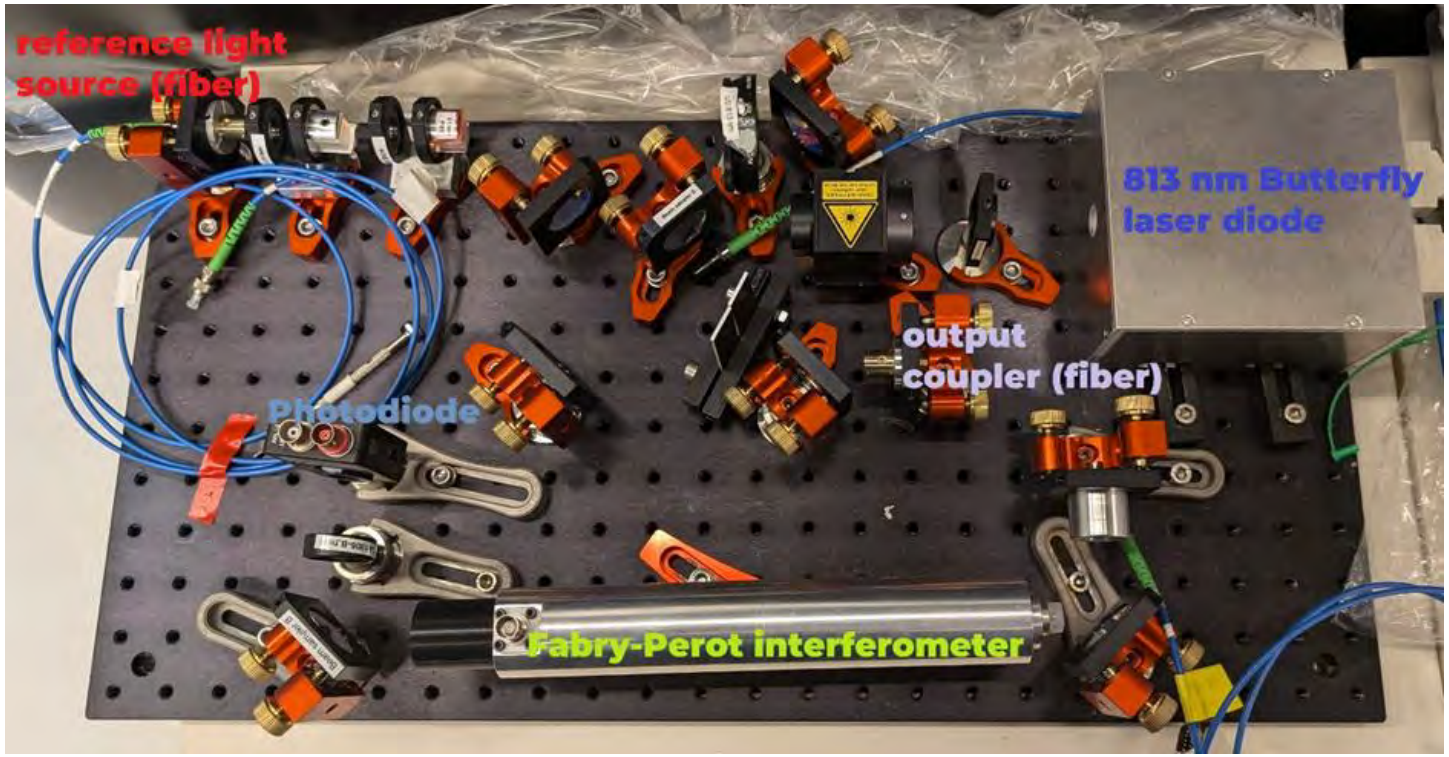
Contacts

Davide Natale, (d.natale@uva.nl)
Junyu He (j.he@uva.nl)
Florian Schreck (f.schreck@uva.nl)

www.strontiumBEC.com

Reviving and characterizing an injection locked laser diode

High precision spectroscopy is a founding block for measurements in fundamental physics. In this project you will revive an optical setup that features a 461-nm and 689-nm laser diode. With a process called injection locking you will force the laser diode to emit in a way smaller frequency range than in its regular operation. You will characterize the emission spectrum with a Fabry-Perot interferometer. This laser will be tuned to the 'magic wavelength' of Sr-88's clock transition. It will be used to trap Strontium ensembles of a couple of μK.



Skills at play: laser physics, atomic physics, laser cooling, python control system development.

Contacts

Andras Gacsbaranyi, (k.a.gacsbaranyi@uva.nl)
Sumit Sarkar (s.sarkar@uva.nl)
Florian Schreck (f.schreck@uva.nl)

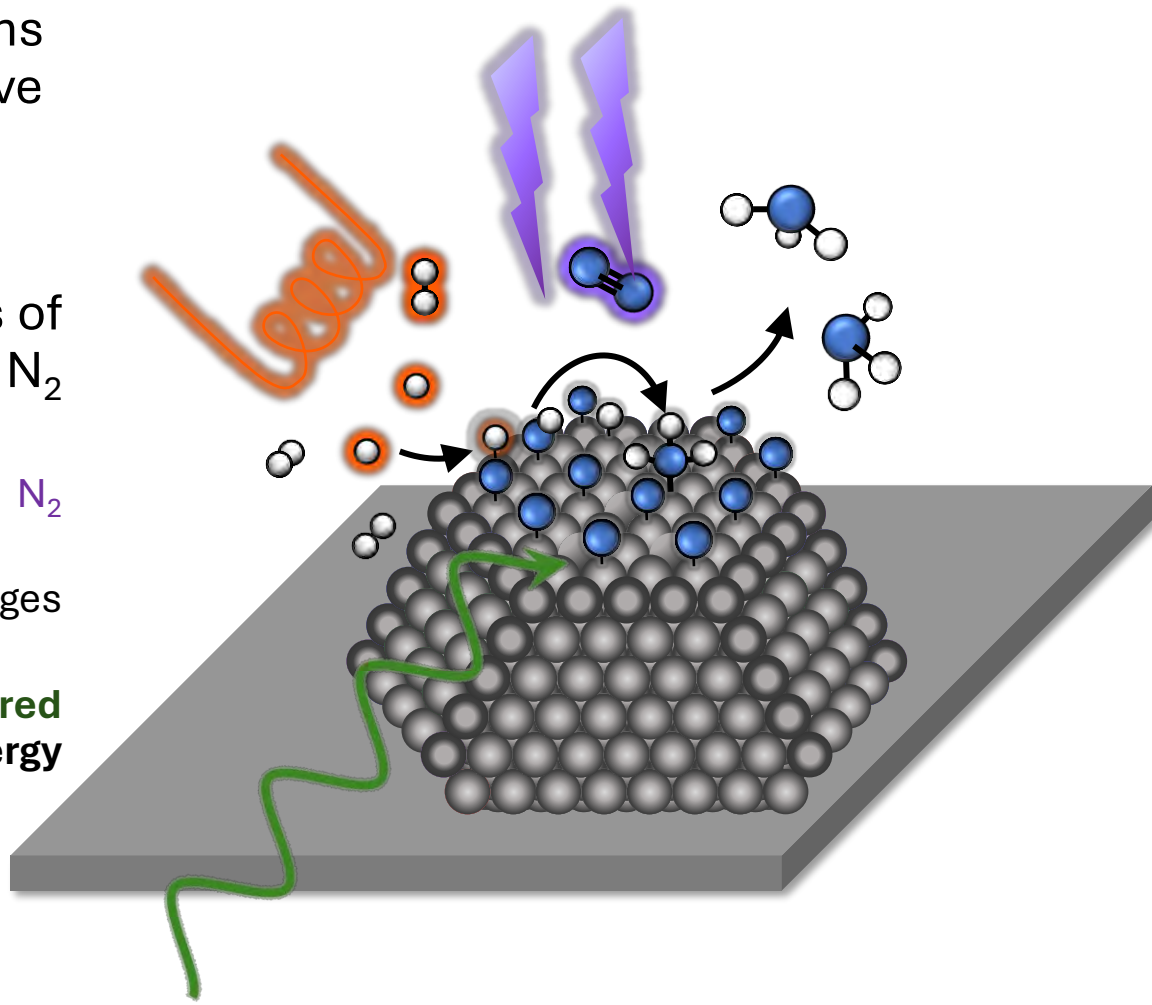
www.strontiumBEC.com

Resolving mechanisms of plasma catalysis using in situ IR spectroscopy

Motivation:
Plasma activation of molecules enhances catalytic reactions at surfaces. The surface reaction mechanisms and the active species are still unknown.

In this project, you will:

- Determine which surface species contribute to reactions of catalytically active metal surfaces in plasma-activated N_2 and H_2.
  - Grow nitride films on single-crystalline metal surfaces, using N_2 plasma in ultra-high vacuum setups
  - Reduce them using atomic hydrogen, during spectroscopy changes in the spectroscopic signature.
  - Characterize their molecular bonds and structure using **infrared reflection absorption spectroscopy (IRRAS)** and **low-energy electron diffraction** (LEED)
  - Monitor products in the gas phase by **mass spectrometry**.



ARCNL

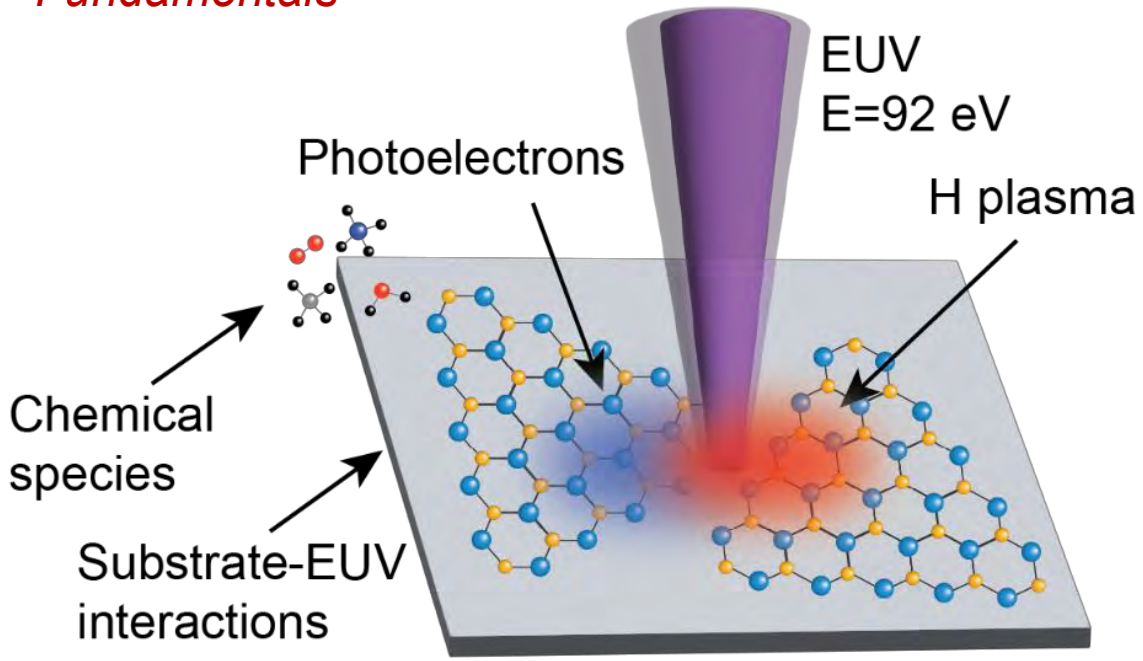
Sabine Auras,
Roland Bliem

Interested? Contact bliem@arcnl.nl

Patterning 2D Semiconductors with EUV light

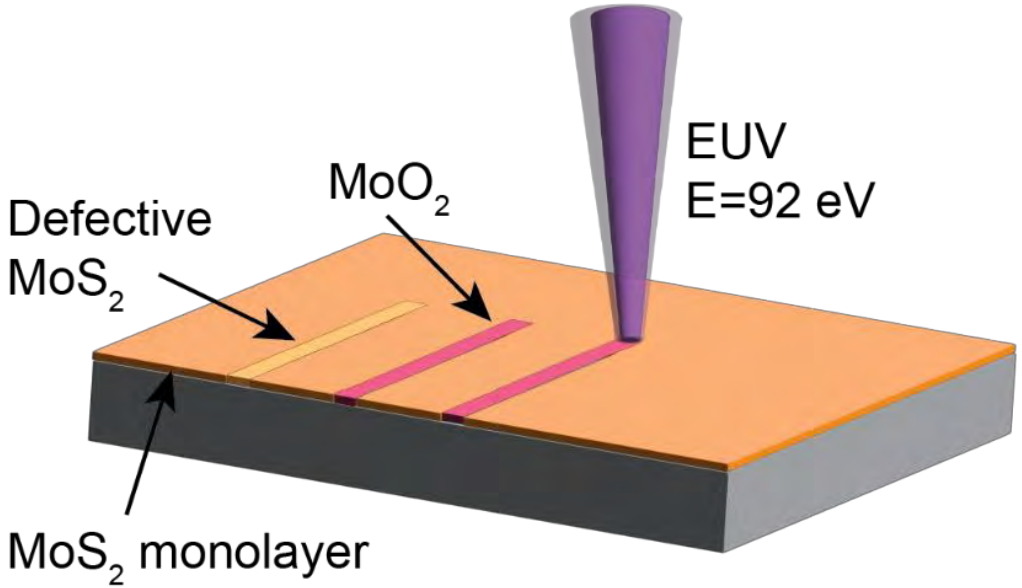
We study fundamental interactions of 2D semiconductors with extreme ultraviolet (EUV) lithography environments to design new patterning approaches for next-generation chips

Fundamentals



EUV lithography parameter space

Patterning approaches



Selective EUV chemical transformations

ARCNL

University of Amsterdam

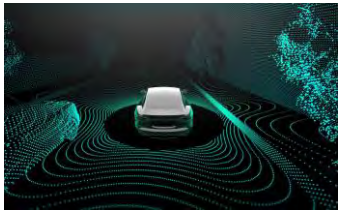
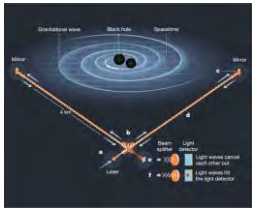
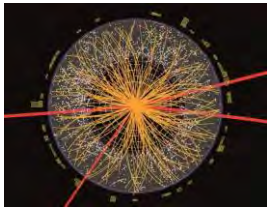
Contact

Dr. Reynolds Dziobek-Garrett

Dr. Roland Bliem (bliem@arcnl.nl)

Collaboration: Dr. Jorik van de Groep

2D material Detectors for high-energy physics

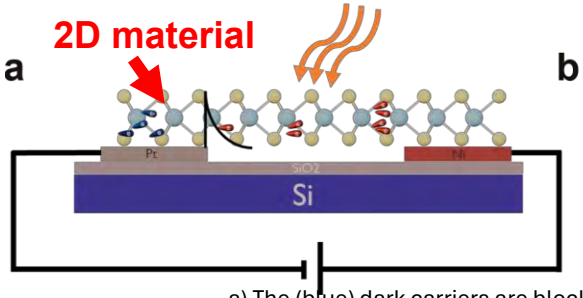




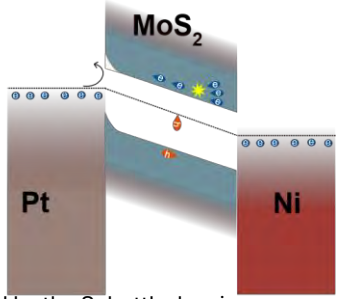
Applications for these detectors, LIDAR, gravitational wave detection and particle physics.

- Future Detector technology: higher sensitivity & time resolution.
- Novel 2D materials promise advanced detectors
- We explore making new 2D material photodetectors for particle detection, dark matter searches and gravitational waves.

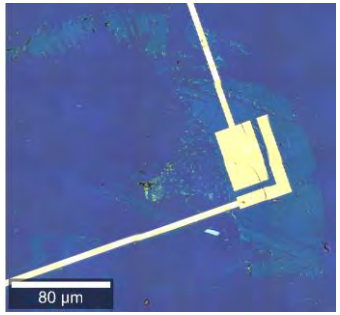
Objective

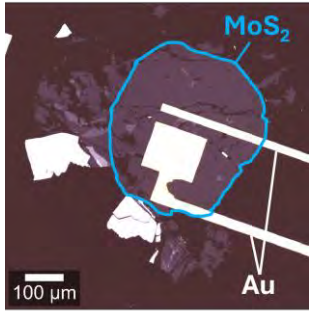
- Make avalanche photodiode using novel 2D materials, like graphene and MoS_2.
- Characterize device using standard particle physics detector instrumentation.
- Research effect of gating on the avalanche process and breakdown voltage





a) The (blue) dark carriers are blocked by the Schottky barrier induced by the platinum contact. b) Band diagram of the device.





Simple prototypes of an APD using Au contacts.

Peter Schall Lab

Prof. Peter Schall, p.schall@uva.nl; Bauke van der Vorm, b.v.vandervorm@uva.nl

The Mystery of Quantum-dots:

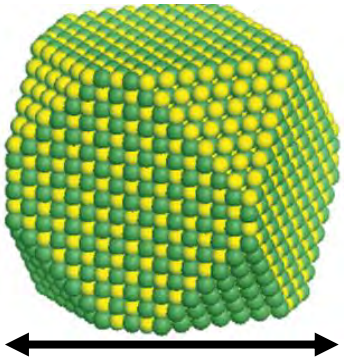
Single photon emission intermittency

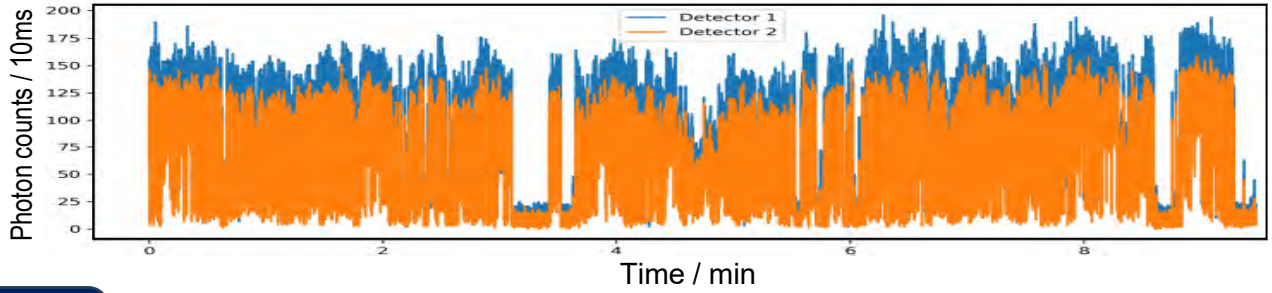
Quantum dots

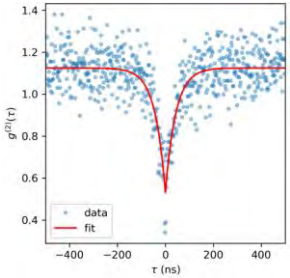
- Quantum light emitters, emitting photons one-by-one
- Show mysterious on-off dynamics (blinking)

Your research

- Directly study single photon emission
- Investigate the mysterious quantum emission intermittency







Peter Schall Lab

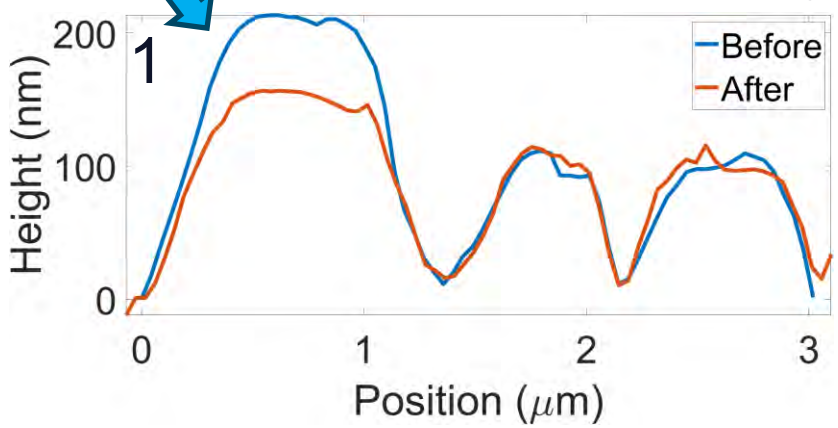
David van Houten (d.vanhouten@uva.nl); Prof. Peter Schall (p.schall@uva.nl)

Diamonds are forever?

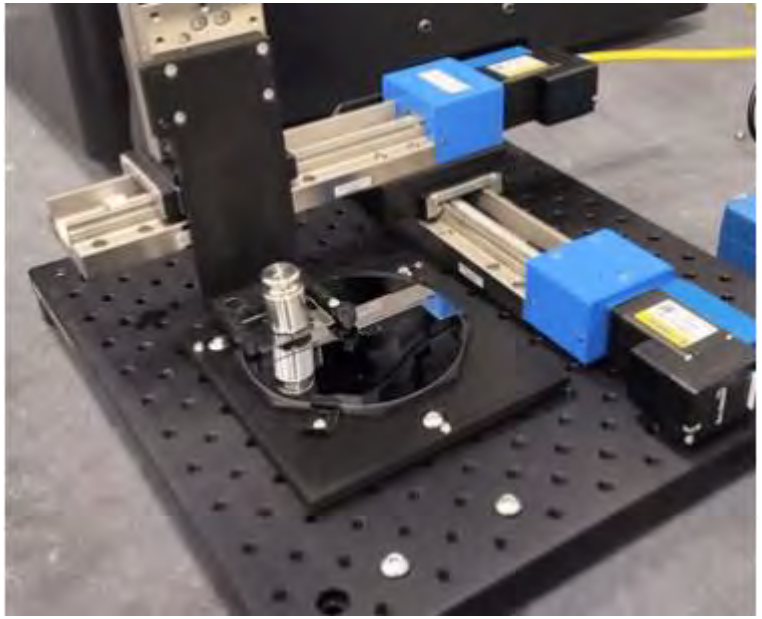
No! Even Diamond wears, when you subject it to $23 \cdot 10^9$ Pa (!) pressure.

What happens at lower contact pressures? You will help us find out in this BSc project.

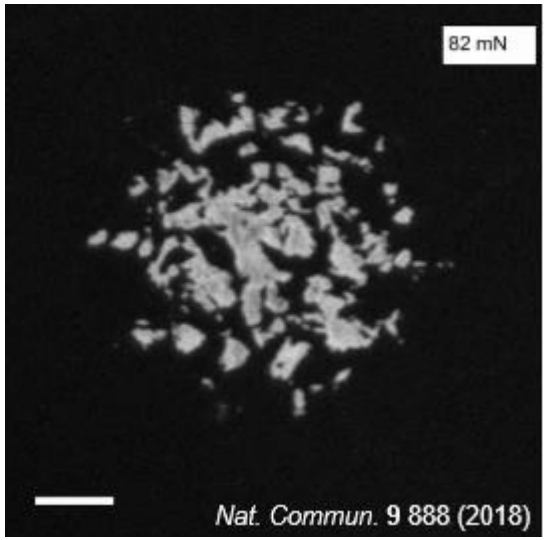
Atomic Force Microscopy



Wear Experiments



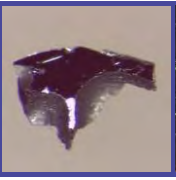
Contact Calculations

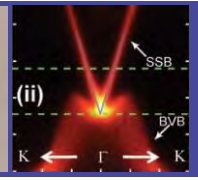


Bart Weber

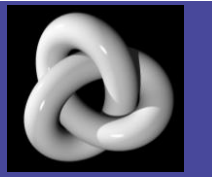
University of Amsterdam

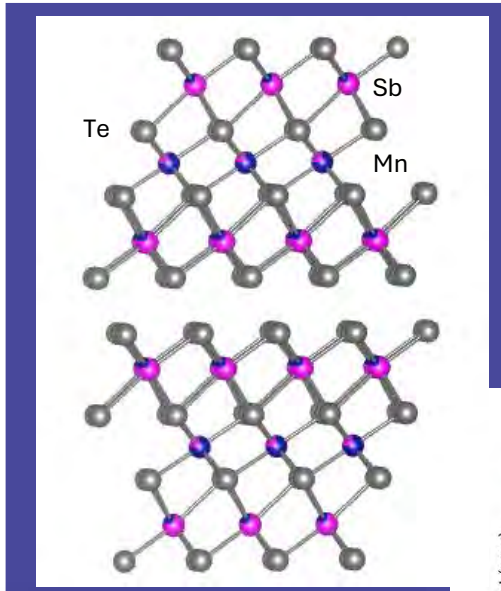
UvA
Van der Waals-Zeeman Institute

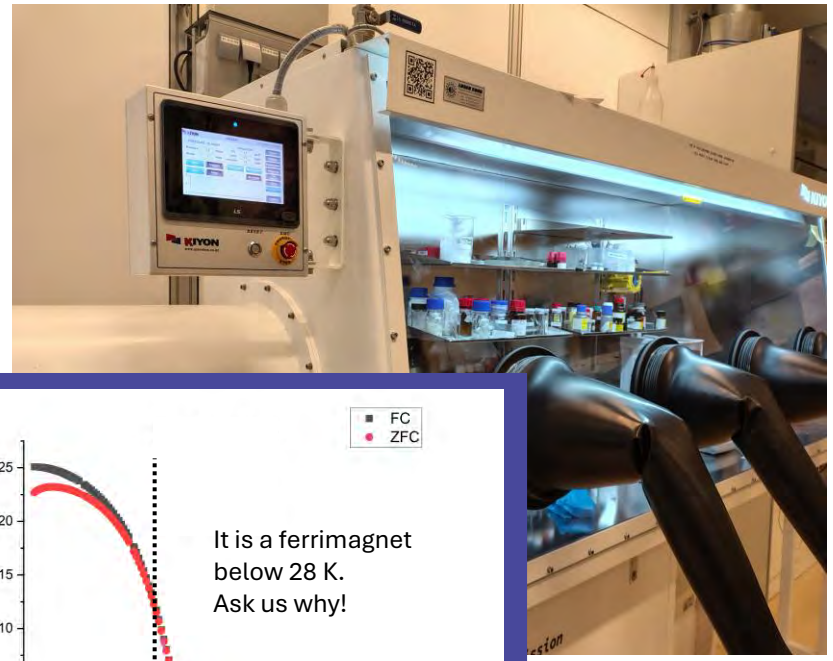


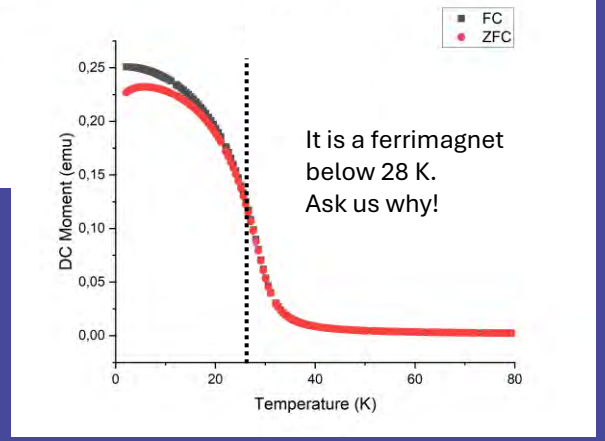


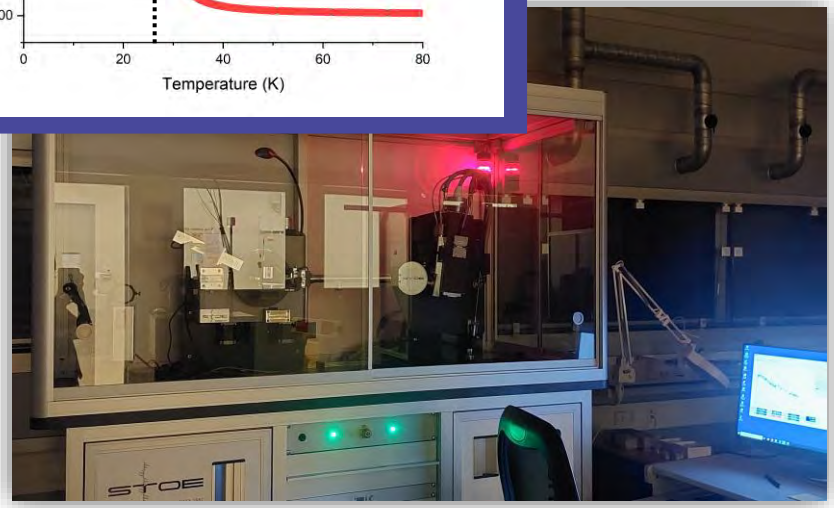
Topological Magnet $MnSb_2Te_4$











Got intrigued by semiconductors and quantum Hall effects in GM1 and QC?

Now you can make your own material: semiconductor + magnetism + quantum = a magnetic topological insulator

We offer:

- *Chance to contribute to the frontier research on quantum materials*
- *Experimental lab work on making polycrystalline materials and measuring and analyzing their magnetic properties*
- *Friendly and enthusiastic team*

Supervisors:
Anna Isaeva (a.isaeva@uva.nl)
Laura Zeni Cunha (l.zenicunha@uva.nl)